SSC CHSL TIER-1 : MASTER REVISION NOTES

Based on Detailed Multi-Year Analysis of Official PYQs (2017 to 2024)
Focus: 100% Guaranteed Common Question Types & Smart Study Rules

SECTION 1 : ENGLISH LANGUAGE (BASIC KNOWLEDGE)

1. Article Rules (Har Shift Me 1 Se 2 Questions Pakke)

Rule 1: Indefinite Article 'A' vs 'An' (Sound Based)
Article 'A' aur 'An' ka use letter dekh kar nahi, balki **Vowel Sound (स्वर ध्वनि: अ, आ, इ, ई, ए, ऐ, ओ, औ)** ke aadhar par hota hai.
• Agar shabd ka pehla uchcharan (pronunciation) Vowel Sound hai $ ightarrow$ **'An'** (e.g., *an umbrella, an American, an honest man, an hour, an heir*).
• Agar uchcharan Consonant Sound (व्यंजन ध्वनि: क, ख, ग, य, व...) hai $ ightarrow$ **'A'** (e.g., *a European* [sound: 'yu'], *a university* [sound: 'yu'], *a one-eyed man* [sound: 'va']).

PYQ Repetition Examples:

• **(CHSL 2024 S2):** "This is a doll and there is a umbrella." $ ightarrow$ **Correction:** *an umbrella* (umbrella starts with 'अ' sound).
• **(CHSL 2024 S2):** "Emily is a American living in Paris." $ ightarrow$ **Correction:** *an American*.
• **(CHSL 2018 S1):** "A stream is unbroken whole..." $ ightarrow$ **Correction:** *an unbroken whole*.

Rule 2: Definite Article 'The' ke 3 Most-Repeated Cases
1. **Superlative Degree:** Superlative adjective se pehle hamesha 'The' lagta hai.
• *PYQ (CHSL 2024 S1):* "This is best book for children." $ ightarrow$ **Correction:** *This is **the best book***.
2. **Phrase "One of the + Plural Noun":**
• *PYQ (CHSL 2020 S1):* "When I met Akbar Padamsee, one of a pioneers..." $ ightarrow$ **Correction:** *one of **the** pioneers*.
3. **Unique / Defined Objects & Holy/Geographical entities:** *The Sun, The Moon, The Himalayas, The Ganga, The Philippines* (islands group).

2. Subject-Verb Agreement & Concord (Most Crucial)

• **Rule A: Singular Subject $ ightarrow$ Singular Verb ($V_1 + s/es$)**
PYQ (CHSL 2024 S2): "It is a general belief that the female ostrich guard the nest..." $ ightarrow$ **Correction:** *female ostrich **guards***.
PYQ (CHSL 2020 S1): "Your skill lie in cheering up people..." $ ightarrow$ **Correction:** *Your skill **lies***.
PYQ (CHSL 2021 S1): "The Prime Minister chair a high-level meeting..." $ ightarrow$ **Correction:** *The Prime Minister **chairs / chaired***.
• **Rule B: Plural Subject $ ightarrow$ Plural Verb**
PYQ (CHSL 2017 S1): "Due to the snow the marks was unrecognisable." $ ightarrow$ **Correction:** *marks **were***.
• **Rule C: "One of the" + Plural Noun + Singular Verb**
PYQ (CHSL 2021 S1): "Public speaking is one of the most feared form..." $ ightarrow$ **Correction:** *most feared **forms***.

3. Adverb vs Adjective Placement Errors

• **Rule:** Verb ko modify karne ke liye **Adverb (-ly form)** lagta hai, Adjective nahi.
PYQ (CHSL 2024 S1): "Passengers started behaving violent when..." $ ightarrow$ **Correction:** *behaving* ***violently***.
PYQ (CHSL 2022 S3): "Rahul sings very sweet when..." $ ightarrow$ **Correction:** *sings* ***very sweetly***.
• **Rule: Placement of 'Enough':**
Adjective ke **baad** 'enough' lagta hai: [Adjective + Enough].
PYQ (CHSL 2024 S2): "The hotel room isn't enough big." $ ightarrow$ **Correction:** *isn't* ***big enough***.

4. Preposition & Conjunction Rules

• **Between vs Among:** 2 ke liye *Between*, 3 ya usse zyada ke liye *Among*.
PYQ (CHSL 2022 S3): "divide the expenses between the three of us" $ ightarrow$ **Correction:** ***among*** *the three of us*.
• **Unless vs Until:** *Unless* = Condition (shart), *Until* = Time (samay). Dono me negative shabd (not) nahi judta.
PYQ (CHSL 2021 S1): "I cannot help you if you tell me your problem." $ ightarrow$ **Correction:** ***unless*** *you tell me*.

5. Active-Passive Voice & Direct-Indirect Speech (100% Fixed Format)

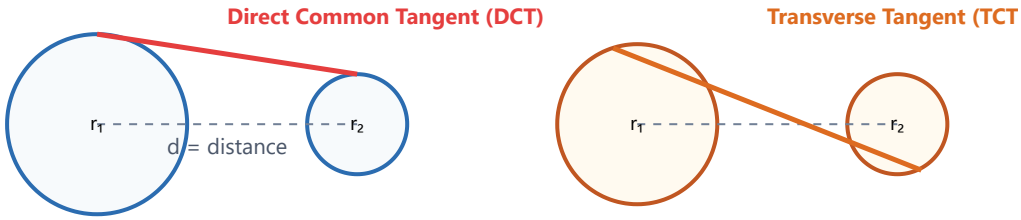
Voice Shortcuts:
• Simple Past (V_2) $ ightarrow$ $was/were + V_3$ [e.g., "*The police beat protestors*" *$ ightarrow$* "*Protestors were beaten*"].
• Imperative (Orders/Offers): "*Celebrate the New Year*" *$ ightarrow$* "***Let*** *the New Year* ***be celebrated***".
Narration Shortcuts:
• Interrogative (Yes/No Question): Reporting verb *said to $ ightarrow$ asked*, connector *if / whether*, sentence assertive ho jata hai.
• Universal Truths / Proverbs: Tense change nahi hota [e.g., *The sage said, "Man proposes and God disposes" $ ightarrow$ The sage said that man proposes and God disposes*].
• Polite Requests: "*Please help me*" *$ ightarrow$ requested + to + V_1*.

6. High Frequency Vocabulary & Spelling Mistakes

Category	Word & Meaning	Exam Trend / Note
OWS (Repeated)	**Kleptomaniac:** Impulse to steal **Misogamist:** Hates marriage **Abdicate:** Renounce throne / power **Emigrant:** Leaves country to settle abroad **Aviary:** Place where birds are kept	Har saal 2 questions OWS aate hain. Common themes: Haters, Lovers, Government types, Living places.
Idioms (Repeated)	• **Cost an arm and a leg:** Very expensive • **At daggers drawn:** Ready to fight / hostile • **Sell like hot cakes:** Sold very quickly • **Up in arms:** Angry and protesting • **Stir up a hornet's nest:** Provoke serious trouble	90% idioms previous year papers se direct repeat hote hain.
Spelling Traps	• **Receive** (not recieve - rule: 'i before e except after c') • **Committee** (double m, double t, double e) • **Argument** (no 'e' after u) • **Conscientious** (sci-en-tious) • **Underrated** (double r)	Hamesha double consonants (mm, tt, rr, ss) aur 'ie/ei' par focus karein.

SECTION 2 : QUANTITATIVE APTITUDE (MATHS FORMULAS & SHORTCUTS)

1. Geometry: Circle Tangents & Chords (Har Paper Ka Favorite)



1. Direct Common Tangent (DCT) Length:

$$L_{DCT} = \sqrt{d^2 - (r_1 - r_2)^2}$$

Yahan d dono kendron (centers) ke bich ki duri hai, aur r_1, r_2 radii hain.

PYQ (CHSL 2023 S1 Q64): $r_1 = 12 ext{ cm}, r_2 = 5 ext{ cm}, d = 25 ext{ cm}$.

$\implies L_{DCT} = \sqrt{25^2 - (12 - 5)^2} = \sqrt{625 - 49} = \sqrt{576} = \mathbf{24 ext{ cm}}$.

2. Transverse Common Tangent (TCT) Length:

$$L_{TCT} = \sqrt{d^2 - (r_1 + r_2)^2}$$

Yahan plus $(r_1 + r_2)$ hota hai kyunki tangent cross karti hai.

PYQ (CHSL 2024 S2 Q53): $r_1 = 7 ext{ cm}, r_2 = 5 ext{ cm}, d = 15 ext{ cm}$.

$\implies L_{TCT} = \sqrt{15^2 - (7 + 5)^2} = \sqrt{225 - 144} = \sqrt{81} = \mathbf{9 ext{ cm}}$.

3. Intersecting Chords Property: Jab do jeevayein (chords) AB aur CD aapas me circle ke andar ya bahar point P par katti hain:

$$PA imes PB = PC imes PD$$

PYQ (CHSL 2020 S1 Q55): Chords PQ aur RS point L par milti hain. $PL = 9, LQ = 4, LS = 6$. Find RL.

$\implies PL imes LQ = RL imes LS \implies 9 imes 4 = RL imes 6 \implies RL = rac{36}{6} = \mathbf{6 ext{ cm}}$.

4. Internal Angle Bisector Theorem (त्रिभुज कोण समद्विभाजक): $ riangle ABC$ me agar AD, $ngle A$ ka bisector hai:

$$rac{AB}{AC} = rac{BD}{DC}$$

PYQ (CHSL 2021 S1 Q74): $BD = 3.6 ext{ cm}, BC = 8 ext{ cm} \implies DC = 8 - 3.6 = 4.4 ext{ cm}$.

$\implies rac{AB}{AC} = rac{3.6}{4.4} = rac{9}{11} \implies \mathbf{9 : 11}$.

5. Similar Triangles Area & Perimeter Ratio:

$$rac{ ext{Area}(riangle ABC)}{ ext{Area}(riangle DEF)} = \left(rac{AB}{DE} ight)^2 = \left(rac{ ext{Perimeter}_1}{ ext{Perimeter}_2} ight)^2$$

PYQ (CHSL 2024 S1 Q62): Perimeters $24 ext{ cm}$ aur $9 ext{ cm}$ hain. Area ratio $= \left(rac{24}{9} ight)^2 = \left(rac{8}{3} ight)^2 = \mathbf{64 : 9}$.

2. Trigonometry Shortcuts & Identities

1. Identity Relation: $\sec^2 heta - an^2 heta = 1$
$\implies (\sec heta + an heta)(\sec heta - an heta) = 1$
$$ ext{If } \sec heta + an heta = k \implies \sec heta - an heta = rac{1}{k} $$
PYQ (CHSL 2023 S1 Q53): If $\sec heta + an heta = 5$, then $\sec heta - an heta = \mathbf{rac{1}{5}}$. (Instant 2 seconds solve!).

2. Minimum Value of Trigonometric Function:
$$ ext{Minimum value of } a\csc^2 heta + b\sin^2 heta = 2\sqrt{ab} $$
PYQ (CHSL 2020 S1 Q68): Find min value of $8\csc^2 heta + 25\sin^2 heta$.
$\implies ext{Min} = 2\sqrt{8 imes 25} = 2\sqrt{200} = 2 imes 10\sqrt{2} = \mathbf{20\sqrt{2}}$.

3. Componendo & Dividendo (C&D) Trick: Agar $rac{\cos heta + \sin heta}{\cos heta - \sin heta} = rac{a}{b}$, toh direct likhein: $\cot heta = rac{a + b}{a - b}$ ya $ an heta = rac{a - b}{a + b}$.
PYQ (CHSL 2020 S1 Q64): $rac{\cos heta + \sin heta}{\cos heta - \sin heta} = 8 = rac{8}{1} \implies \cot heta = rac{8 + 1}{8 - 1} = \mathbf{rac{9}{7}}$.

3. Algebra Must-Know Formulas

• **Master Formula 1:** Agar $a + b + c = 0$, tab:
$$ a^3 + b^3 + c^3 - 3abc = 0 \iff a^3 + b^3 + c^3 = 3abc $$
PYQ (CHSL 2023 S1 Q51): Direct formula pucha gaya tha $\implies \mathbf{0}$.
• **Master Formula 2:** $(a+b)^2 - 4ab = (a-b)^2$
PYQ (CHSL 2022 S3 Q67): $(957 + 932)^2 - 4 imes 957 imes 932 = (957 - 932)^2 = (25)^2 = \mathbf{625}$.
• **Master Formula 3:** If $x + rac{1}{x} = k \implies x^3 + rac{1}{x^3} = k^3 - 3k$.

4. Divisibility Rules (Number System)

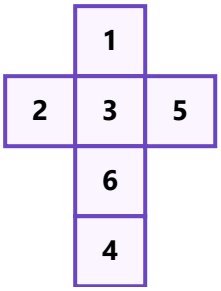
Divisor	Condition / Trick	PYQ Example
Divisible by 3 / 9	Digit sum 3 ya 9 se kate.	**(2024 S1):** $627x5$ divisible by 9 $ ightarrow 6+2+7+x+5 = 20+x \implies x=\mathbf{7}$.
Divisible by 11	(Sum of odd places) - (Sum of even places) = 0 ya 11 ka multiple.	**(2024 S2):** $3c2933k$ divisible by 5 and 11 $\implies k=5, c=1 \implies c+k=\mathbf{6}$.
Divisible by 72 / 24	Co-primes me todein: $72 = 8 imes 9$, $24 = 3 imes 8$. Last 3 digits 8 se katein aur digit sum 9/3 se kate.	**(2022 S3):** 478656 is divisible by 24 (digit sum = 36).

5. Mensuration (2D & 3D Volume / Melting Formulas)

• **Sphere (गोला):** Volume $= rac{4}{3}\pi r^3$, Surface Area $= 4\pi r^2$.
• **Melting & Recasting Trick:** Badha volume = $N imes$ Chota volume.
PYQ (CHSL 2021 S1 Q70): Sphere $R = 10 ext{ cm}$ ko pighla kar $r = 2 ext{ cm}$ ke kitne gole banenge?
$\implies N = rac{rac{4}{3}\pi (10)^3}{rac{4}{3}\pi (2)^3} = \left(rac{10}{2} ight)^3 = 5^3 = \mathbf{125}$.
• **Cube se Sphere nikalna:** Side a ke cube se sabse bada sphere $\implies ext{Radius } r = rac{a}{2}$.

SECTION 3 : GENERAL INTELLIGENCE & REASONING

1. Open Dice & Closed Dice (1 Question Pakka)



Universal Open Dice Rule: Alternate Boxes Opposite Hote Hain
• Ek seedhi line me ek chhodkar agla dabba (alternate face) opposite hota hai.
• Upar diye figure me:
- Vertical line me: $1 \leftrightarrow 6$ aur $3 \leftrightarrow 4$
- Baaki bache do side dabbey: $2 \leftrightarrow 5$
• **Golden Elimination Trick:** Jo do number ek dusre ke opposite hote hain, wo kabhi bhi fold hone par **saath me (adjacent) nahi dikh sakte**. Agar option me saath dikh rahe hain, toh wo option turant galat hai!

2. Mathematical Operations (BODMAS Sign Balancing)

Solving Strategy (Speed Trick):
1. Hamesha **Divide ($\div$) sign** par pehle nazar daalo. Agar divide sign aisi do sankhyao ke beech aa raha hai jo poori tarah divide nahi hoti (e.g., $14 \div 5$), toh wo option turant eliminate ho jayega (jab tak right side me fraction na ho).
2. *PYQ (CHSL 2022 S3 Q34):* $(13 \times 9) \div 3 + (14 \div 7) \times 6 - 21 + (32 \times 2) \Rightarrow 39 + 12 - 21 + 64 = \mathbf{94}$.

3. Mirror Image & Water Image

• **Mirror on Vertical line (MN / AB at Right or Left):** Left becomes Right, and Right becomes Left. (Top aur Bottom same rehte hain).
• **Mirror at Bottom (Water Image):** Top becomes Bottom, and Bottom becomes Top. (Left aur Right same rehte hain).
• **Quick Elimination:** Sabse pehle pehle aur aakhiri letter ka inversion check karo, 4 me se 2-3 options turant cut jaate hain.

4. Set Analogy Patterns (Most Common SSC Logics)

Pattern Type	Formula Logic	Real PYQ Example
Sum ka Multiple / Square	$((A + B) / 2)^2 = C$	**(2021 S1):** $(12, 18, 225) \implies ((12+18)/2)^2 = 15^2 = 225$.
Square Sum Multiple	$(A^2 + C^2) imes 2 = B$	**(2020 S1):** $(3, 68, 5) \implies (3^2 + 5^2) imes 2 = (9+25) imes 2 = 68$.
Difference of Squares	$A^2 - B^2 = C$	**(2021 S1):** $(7, 4, 33) \implies 7^2 - 4^2 = 49 - 16 = 33$.
Cube + Square	$A^3 + C^2 = B$	**(2022 S3):** $(3, 43, 4) \implies 3^3 + 4^2 = 27 + 16 = 43$.

SECTION 4 : GENERAL AWARENESS (POLITY, ART & CULTURE, SCIENCE)

1. Indian Constitution: All Important Articles (100% Guaranteed Topic)

Article No.	Subject / Provision	Why It Appears in SSC
Art 14	Equality before law (विधि के समक्ष समता)	Fundamental Right (Part III)
Art 15	Prohibition of discrimination on grounds of religion, race, caste, sex, birth place	CHSL 2022 Shift-3 direct question
Art 19	6 Freedoms (Speech, Assembly, Association, Movement, Residence, Profession)	Most tested fundamental right
Art 20 & 22	Protection against conviction (Art 20) & Protection against arrest/detention (Art 22)	CHSL 2022 Shift-3 direct question
Art 21 & 21A	Right to Life (Art 21) & Right to Free Elementary Education (Art 21A - 86th Amd)	Frequently asked in all SSC exams
Art 23 & 24	Right against Exploitation: Human trafficking prohibited (23), Child labour banned (24)	Very common in Tier-1
Art 32	Constitutional Remedies (Supreme Court Writs) - "Heart and Soul of Constitution"	Said by Dr. B.R. Ambedkar
Art 40	Organisation of Village Panchayats (DPSP)	Connects with 73rd Amendment
Art 44	Uniform Civil Code (समान नागरिक संहिता)	Hot current topic & repeat PYQ
Art 51A	**Fundamental Duties (मौलिक कर्तव्य)**: 10 duties added by 42nd Amd (1976), 11th duty added by 86th Amd (2002)	**CHSL 2024 & 2023 dono me aaya!**
Art 93	Lok Sabha Speaker & Deputy Speaker	CHSL 2020 Shift-1 direct question
Art 124 to 147	Supreme Court of India (Jurisdiction, Advisory powers under Art 143)	Canada se li gayi Advisory jurisdiction
Art 243 to 243O	Panchayati Raj (Part IX, 11th Schedule - 29 subjects)	73rd Amendment 1992 (CHSL 2018)
Art 352, 356, 360	Emergency Provisions (National, President's Rule, Financial)	Core Polity syllabus
Art 355	Duty of Union to protect States against external aggression & internal disturbance	CHSL 2023 Shift-1 direct question
Art 368	Power of Parliament to Amend Constitution (Part XX)	CHSL 2017 & 2021 direct question

2. Major Constitutional Amendments & Sources Borrowed

• **42nd Amendment (1976):** Known as "Mini Constitution". Added words *Socialist, Secular, Integrity* to Preamble. Added Fundamental Duties (Part IVA, Art 51A) on recommendation of Swaran Singh Committee.
• **44th Amendment (1978):** Right to Property removed from Fundamental Rights and made legal right (Art 300A).
• **73rd & 74th Amendments (1992):** Panchayati Raj (Part IX, 11th Schedule) & Municipalities (Part IXA, 12th Schedule).
• **86th Amendment (2002):** Added Article 21A (Right to Education) and 11th Fundamental Duty.
• **Borrowed Features:** Canada $ ightarrow$ Advisory jurisdiction of Supreme Court, Quasi-federal structure; USSR $ ightarrow$ Fundamental Duties, Five-Year Plans; UK $ ightarrow$ Parliamentary system, Rule of Law; USA $ ightarrow$ Fundamental Rights, Judicial Review.

3. Art & Culture: Classical Dances, Instruments & Masters

Field / Art	Famous Personality / Legend	State / Details
Santoor	**Pandit Shiv Kumar Sharma**, Rahul Sharma, Bhajan Sopori, Satish Vyas	Kashmir traditional instrument (Santoor)
Shehnai	**Ustad Bismillah Khan** (Played at Red Fort on 15 Aug 1947)	Bharat Ratna awardee
Sitar	**Pandit Ravi Shankar**	Hindustani classical instrumental, Bharat Ratna
Sarangi	**Ustad Bundu Khan**, Ram Narayan	Classical string instrument (CHSL 2022)
Kathak	**Birju Maharaj, Shambhu Maharaj**, Lachhu Maharaj	Lucknow Gharana, Uttar Pradesh
Odissi	**Kelucharan Mohapatra, Sanjukta Panigrahi**	Odisha (Kelucharan also played Pakhawaj/Tabla)
Bharatanatyam	**Rukmini Devi Arundale** (1st dancer in Rajya Sabha), Padma Subrahmanyam	Tamil Nadu
Kuchipudi	**Mallika Sarabhai**, Yamini Krishnamurthy	Andhra Pradesh
Folk Paintings	**Warli** (Maharashtra), Madhubani (Bihar), Pattachitra (Odisha)	Tribal / Folk art forms
State Festivals	**Saga Dawa, Losoong, Sakewa** (Sikkim); **Nehru Boat Race** (Kerala)	North-East & South India festivals

4. Geography & Environment Essentials

• **Tropic of Cancer (23°30' N):** Passes through 8 Indian States $ ightarrow$ **Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram** (Mnemonic: *G-R-M-C-J-W-T-M*).
• **Atmosphere Layers:**
1. *Troposphere (क्षोभमंडल):* Lowest layer, avg height 13 km, all weather events occur here.
2. *Stratosphere (समतापमंडल):* Contains Ozone (O_3) layer, ideal for flying airplanes.
3. *Mesosphere (मध्यमंडल):* Coldest layer (up to 80 km), meteorites (उल्कापिंड) burn here.
4. *Thermosphere / Ionosphere:* Radio transmission.
• **Major Sea Ports:** West Coast: *Kandla/Deendayal (Gujarat), Mormugao (Goa), JNPT (Maharashtra)*. East Coast: *Haldia (WB), Paradip (Odisha), Tuticorin/V.O. Chidambaranar (TN)*.

5. General Science & Computer Basics

Topic	Core Key Points / Formulas	PYQ Reference
Human Body	• **Stomach Acid:** HCl (pH 1.5 - 3.0) activates Pepsin for protein digestion. • **Insulin:** Secreted by Pancreas beta cells; deficiency causes Diabetes Mellitus. • **ANF (Atrial Natriuretic Factor):** Secreted by Heart atrial wall to regulate blood pressure.	CHSL 2022 S3 CHSL 2020 S1
Chemistry	• **Galvanisation:** Applying a zinc coating to iron/steel to prevent rusting. • **Law of Constant Proportion:** Joseph Proust (1806). • **Bakelite:** Thermosetting plastic, poor conductor of heat and electricity.	CHSL 2020 S1 CHSL 2022 S3 CHSL 2017 S1
Physics	• $KE = ↑rac{1}{2}mv^2$ | $1 ext{ Joule} = 1 ext{ N}\cdot ext{m}$. • Electric Field: First introduced by Michael Faraday. • Refractive Index: $n = ↑rac{c}{v}$ ($c = ext{speed of light in vacuum}$).	CHSL 2018 S1 CHSL 2021 S1 CHSL 2022 S3
Computer Basics	• **MS Excel 365:** Close workbook shortcut $ ightarrow$ **Ctrl + W**. • Excel grid: Rows are identified by **Numbers** ($1, 2, 3\dots$), Columns by **Letters** ($A, B, C\dots$). • **Average Function:** Present under the **AutoSum** drop-down menu. • **ASCII:** American Standard Code for Information Interchange (128 characters).	CHSL 2023 S1 CHSL 2024 S2 CHSL 2024 S1 CHSL 2021 S1